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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning (AIML)

Assignment Practical No. 6

Adventure Works Sales & Profitability Dashboard

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Course	Data Visualization Lab (TE7663)
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PUBLISH LINK

https://github.com/ManavG27/DATA-VIZUALIZATION-BLOG/blob/main/Assignment_Practical_6_Manav_Goel-1.pdf

1. Aim and Objectives

This practical aims to design analytical Key Performance Indicators (KPIs) and custom DAX calculations in Power BI Desktop, to validate them using DAX Studio, and to present the results through an interactive dashboard. The analysis addresses the following question: which products, categories and regions contribute most to the sales and profit of Adventure Works between January 2020 and June 2022, and how does performance change over time?

The objectives of the study are as follows:

- To create the ten required KPI measures.
- To implement at least twenty custom DAX measures (twenty-five in total).
- To perform time-based analysis covering monthly sales, previous-period sales, sales variance and sales growth percentage.
- To execute and validate DAX queries in DAX Studio and record the results.
- To develop an interactive dashboard comprising KPI cards, at least four visualizations, and slicers for year, product category and region.
- To analyze the results and report the key insights and conclusions.

2. Dataset Description and Data Preparation

The Adventure Works dataset was used, and only the tables required for the analysis were loaded into the model. The data spans 1 January 2020 to 30 June 2022; consequently, the year 2022 represents a half-year period.

Table	Rows	What it holds
Sales (2020–2022 combined)	56,046	Every order line: date, order number, product, customer, territory, quantity
Products	293	Product names, costs and prices
Subcategories	37	Connects products to categories
Categories	4	Bikes, Components, Clothing, Accessories
Territories	10	Region, country and continent
Customers	18,148	Customer details (after cleaning)
Dates	912	Calendar from 1 Jan 2020 to 30 Jun 2022

Data cleaning. The three yearly sales files were combined into a single Sales table using the Folder connector in Power Query. Six non-customer records were removed from the Customers file (three rows with a key of the form “30---”, one blank row and two export-note lines), and the CustomerKey column was converted to the Whole Number data type. This resolved the duplicate-null error and enabled the relationship between the Sales and Customers tables.

Calculated columns. The sales files contain order quantities only. Sales and cost amounts were therefore derived by multiplying the quantity by the price and cost of the corresponding product:

```
Sales Amount = Sales[OrderQuantity] * RELATED(Products[ProductPrice])
Cost Amount  = Sales[OrderQuantity] * RELATED(Products[ProductCost])
```

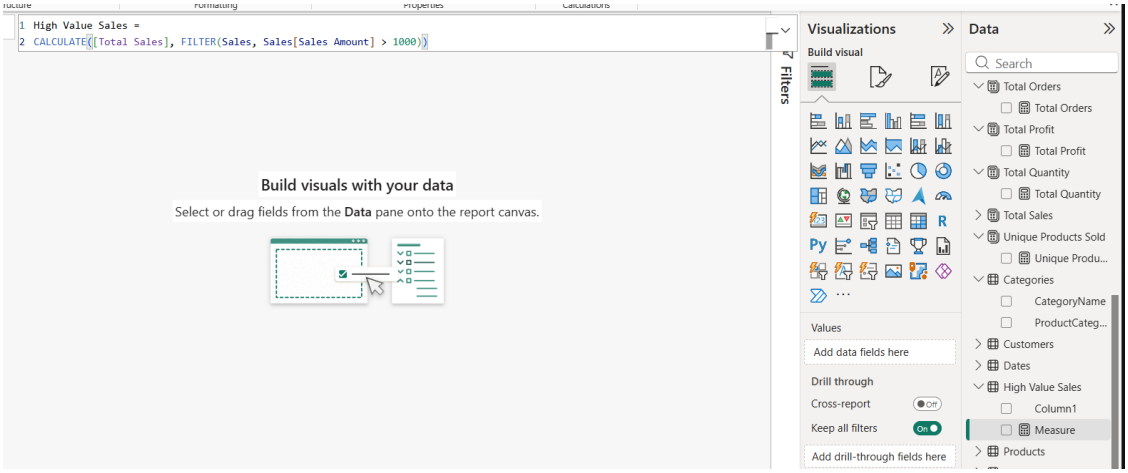
Data model. All relationships in the model are many-to-one with a single filter direction: Sales to Products, Customers, Territories and Dates, and Products to Subcategories to Categories. The Dates table contains the Year, Month Number, Month Name and YearMonth columns and is designated as the date table so that time-intelligence functions operate correctly.

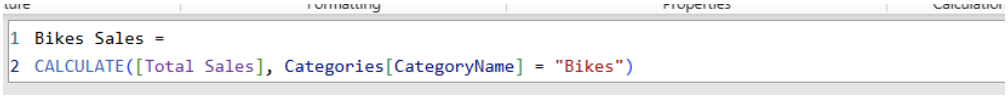
3. Design of KPI Measures and Custom DAX Calculations

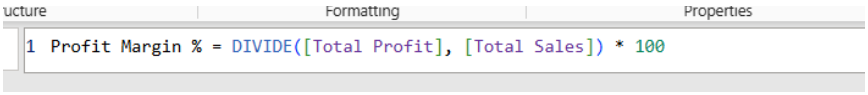
A total of twenty-five measures were created in a dedicated Measures table. These comprise the ten required KPIs, eleven additional measures employing functions such as CALCULATE, FILTER and RANKX, and four time-based measures.

Measure	DAX	What it gives
A. Required KPI measures (10)		
Total Sales	`SUM(Sales[Sales Amount])`	Total sales amount
Total Cost	`SUM(Sales[Cost Amount])`	Total product cost
Total Profit	`[Total Sales] - [Total Cost]`	Sales minus cost
Total Quantity	`SUM(Sales[OrderQuantity])`	Units sold
Total Orders	`DISTINCTCOUNT(Sales[OrderNumber])`	Number of orders
Total Customers	`DISTINCTCOUNT(Sales[CustomerKey])`	Unique customers
Average Sales	`AVERAGE(Sales[Sales Amount])`	Average per order line
Average Order Value	`DIVIDE([Total Sales], [Total Orders])`	Sales per order
Average Product Price	`AVERAGE(Products[ProductPrice])`	Average list price
Profit Margin %	`DIVIDE([Total Profit], [Total Sales]) * 100`	Profit as % of sales
B. Additional measures (11)		
Order Line Items	`COUNT(Sales[OrderLineItem])`	Number of order lines
Sales per Customer	`DIVIDE([Total Sales], [Total Customers])`	Revenue per customer
Profit per Order	`DIVIDE([Total Profit], [Total Orders])`	Profit per order
Avg Quantity per Order	`DIVIDE([Total Quantity], [Total Orders])`	Units per order
Unique Products Sold	`DISTINCTCOUNT(Sales[ProductKey])`	How many products sold
Bikes Sales	`CALCULATE([Total Sales], Categories[CategoryName] = "Bikes")`	Sales from Bikes only
High Value Sales	`CALCULATE([Total Sales], FILTER(Sales, Sales[Sales Amount] > 1000))`	Sales from lines over 1,000
Sales % of Total	`DIVIDE([Total Sales], CALCULATE([Total Sales], ALLSELECTED())) * 100`	Share of selected total
Product Rank	`RANKX(ALL(Products[ProductName]), [Total Sales], , DESC, DENSE)`	Product ranking
Region Rank	`RANKX(ALL(Territories[Region]), [Total Sales], , DESC, DENSE)`	Region ranking
Category Rank	`RANKX(ALL(Categories[CategoryName]), [Total Sales], , DESC, DENSE)`	Category ranking
C. Time-based measures (4)		
Monthly Sales	`CALCULATE([Total Sales], DATESMTD(Dates[Date]))`	Sales for the month
Previous Month Sales	`CALCULATE([Total Sales], DATEADD(Dates[Date], -1, MONTH))`	Last month's sales
Sales Variance	`[Total Sales] - [Previous Month Sales]`	Change from last month
Sales Growth %	`DIVIDE([Sales Variance], [Previous Month Sales]) * 100`	Month-on-month growth

Illustrative Screenshots of DAX Measure Creation

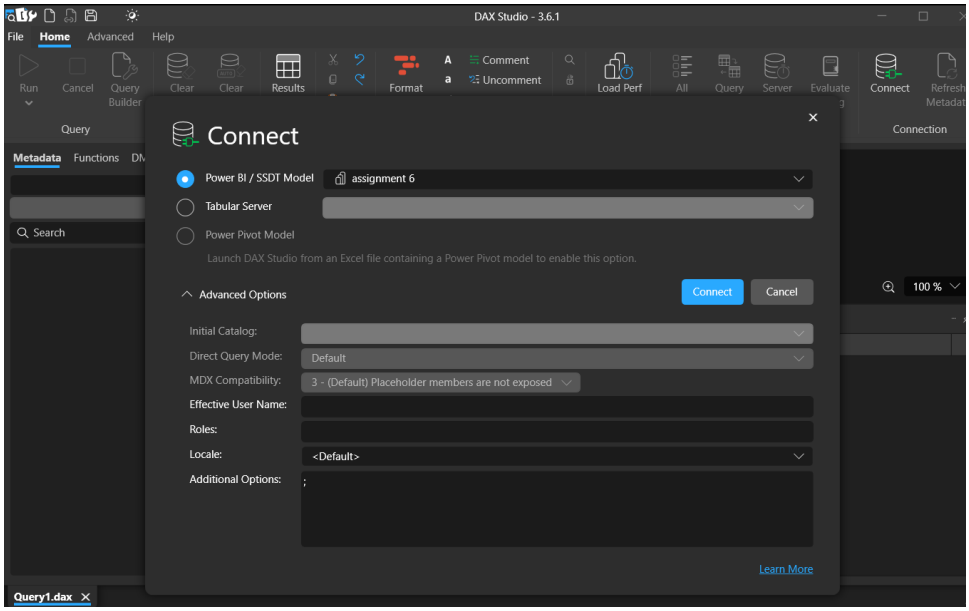






4. Validation Using DAX Studio

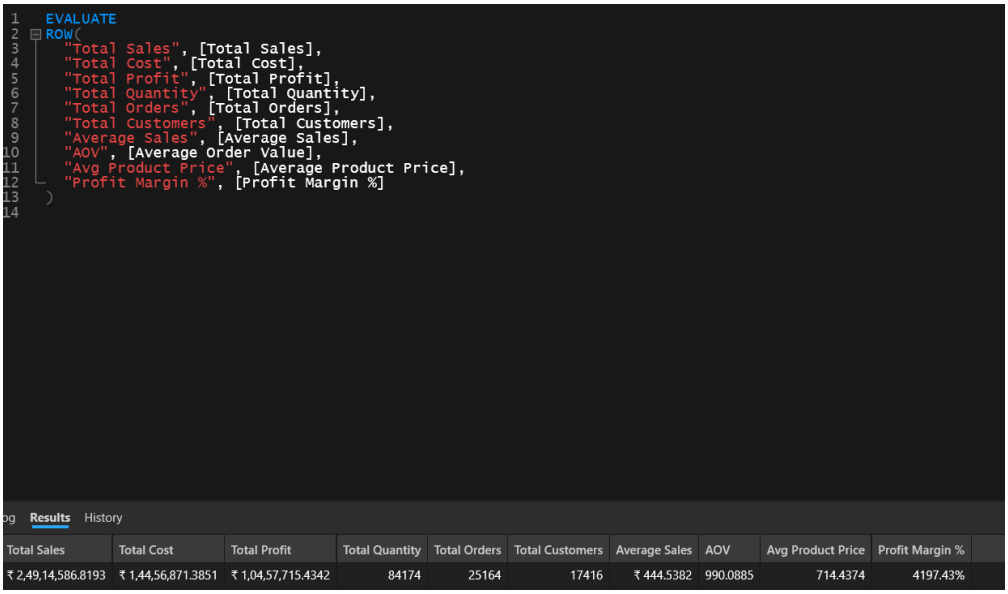
Because the External Tools tab was not available in the Power BI ribbon, DAX Studio was launched independently and connected to the open Power BI file through the PBI / SSDT Model option. Queries could then be executed directly against the data model.



Query 1: Validation of KPI Measures

This query returns all required KPI measures in a single row so that they can be compared with the values displayed on the dashboard cards.

```
EVALUATE
ROW(
  "Total Sales", [Total Sales],
  "Total Cost", [Total Cost],
  "Total Profit", [Total Profit],
  "Total Quantity", [Total Quantity],
  "Total Orders", [Total Orders],
  "Total Customers", [Total Customers],
  "Average Sales", [Average Sales],
  "AOV", [Average Order Value],
  "Avg Product Price", [Average Product Price],
  "Profit Margin %", [Profit Margin %]
)
```

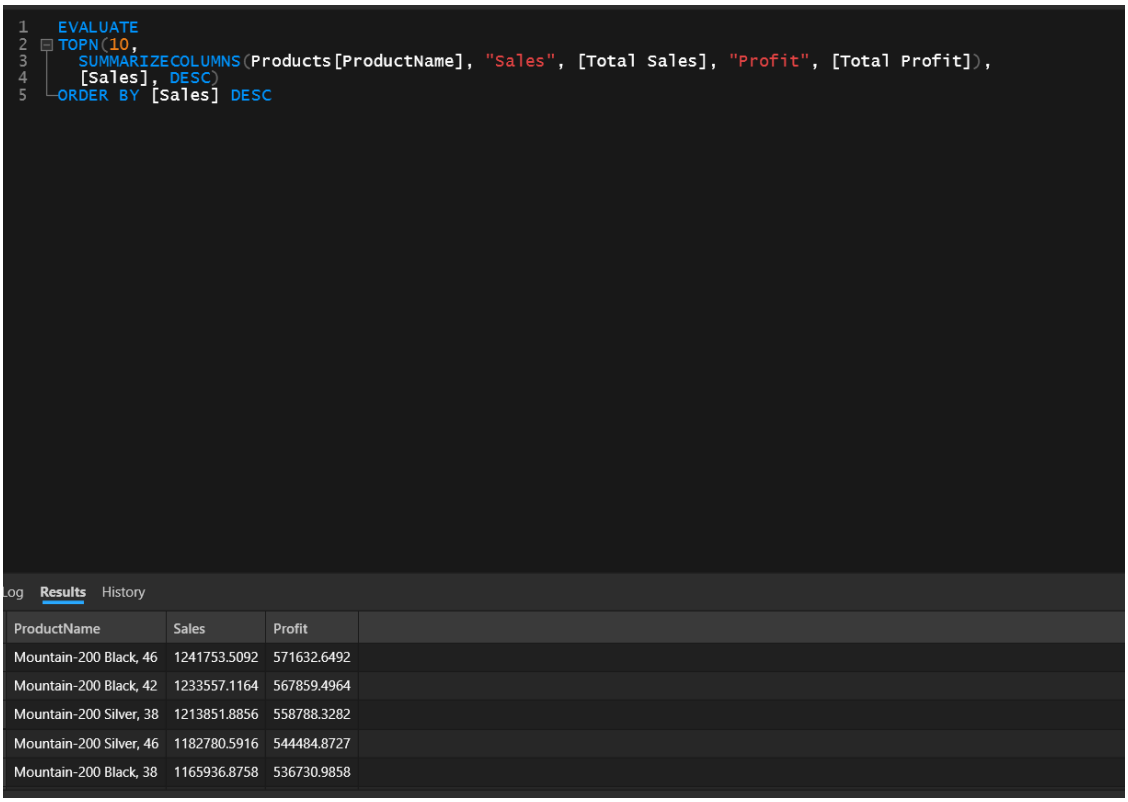


The same totals were also computed directly from the raw CSV files, independently of Power BI. All values agreed with the model output:

Measure	Value	Matches Raw Data?
Total Sales	24,914,586.82	Yes
Total Cost	14,456,871.39	Yes
Total Profit	10,457,715.43	Yes
Total Quantity	84,174	Yes
Total Orders	25,164	Yes
Total Customers	17,416	Yes
Average Sales (per line)	444.54	Yes
Average Order Value	990.09	Yes
Average Product Price	714.44	Yes
Profit Margin %	41.97	Yes

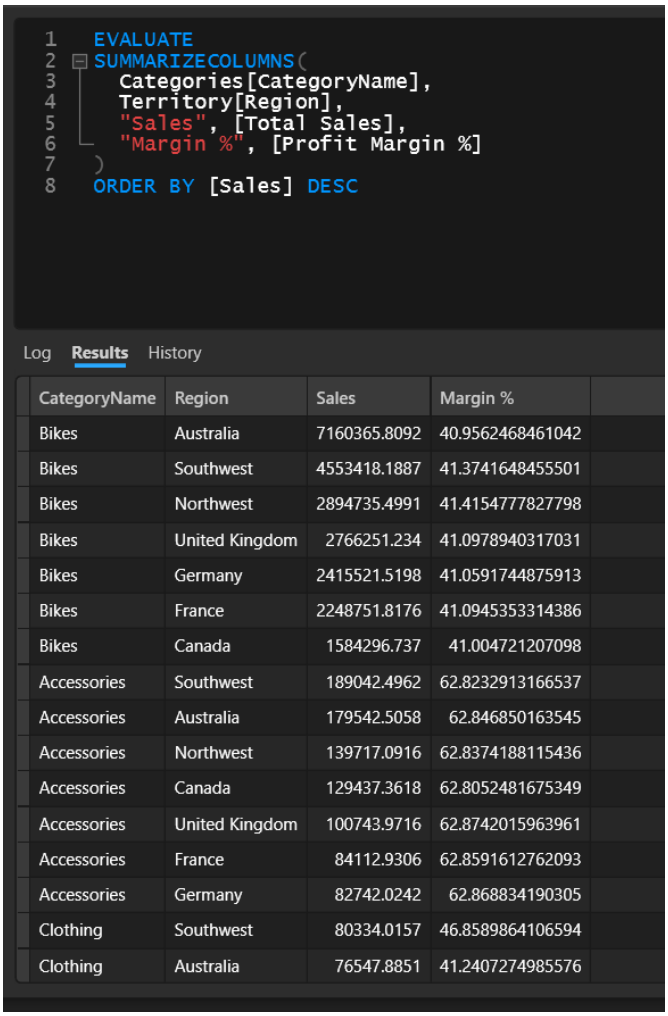
Query 2: Top Ten Products by Sales

```
EVALUATE
TOPN(10, SUMMARIZECOLUMNS(Products[ProductName], "Sales", [Total Sales],
"Profit", [Total Profit]),
  [Sales], DESC)
ORDER BY [Sales] DESC
```



Query 3: Sales and Profit Margin by Category and Region

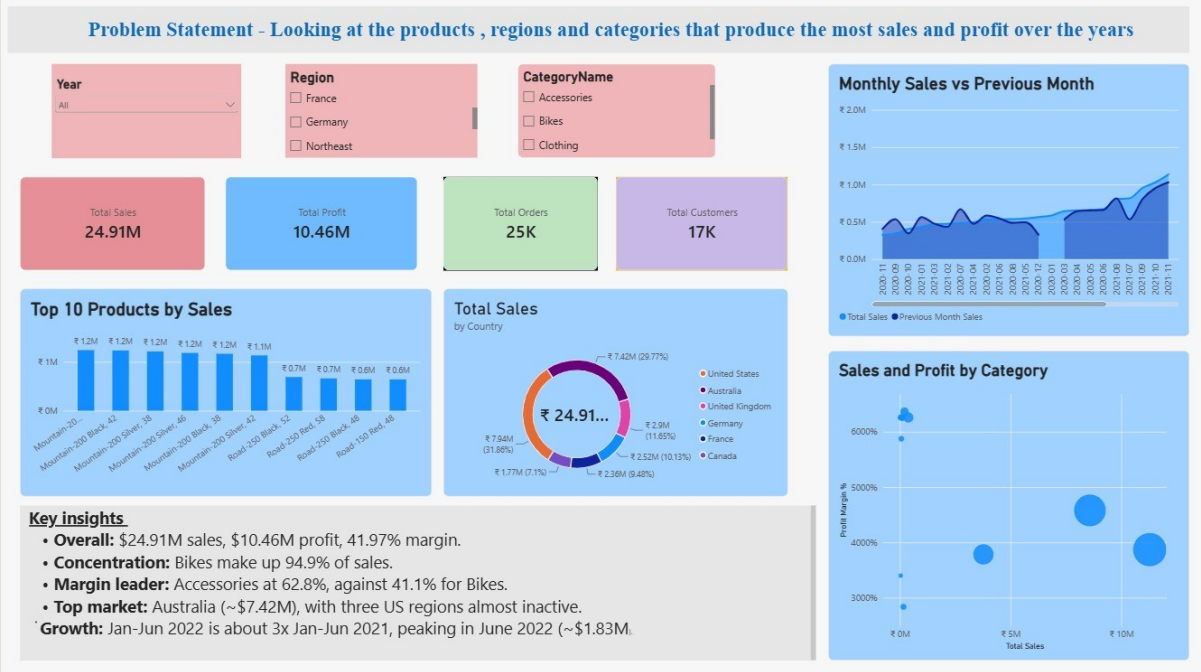
```
EVALUATE
SUMMARIZECOLUMNS(Categories[CategoryName], Territories[Region],
  "Sales", [Total Sales], "Margin %", [Profit Margin %])
ORDER BY [Sales] DESC
```



CategoryName	Region	Sales	Margin %
Bikes	Australia	7160365.8092	40.9562468461042
Bikes	Southwest	4553418.1887	41.3741648455501
Bikes	Northwest	2894735.4991	41.4154777827798
Bikes	United Kingdom	2766251.234	41.0978940317031
Bikes	Germany	2415521.5198	41.0591744875913
Bikes	France	2248751.8176	41.0945353314386
Bikes	Canada	1584296.737	41.004721207098
Accessories	Southwest	189042.4962	62.8232913166537
Accessories	Australia	179542.5058	62.846850163545
Accessories	Northwest	139717.0916	62.8374188115436
Accessories	Canada	129437.3618	62.8052481675349
Accessories	United Kingdom	100743.9716	62.8742015963961
Accessories	France	84112.9306	62.8591612762093
Accessories	Germany	82742.0242	62.868834190305
Clothing	Southwest	80334.0157	46.8589864106594
Clothing	Australia	76547.8851	41.2407274985576

5. Interactive Power BI Dashboard

The dashboard comprises four KPI cards (Total Sales, Total Profit, Total Orders and Total Customers), slicers for Year, Region and CategoryName, four visualizations and a key-insights panel. Selecting a value in any slicer dynamically updates every card and chart on the page.



Visual	Fields	Purpose
Area / line chart	X: Year-Month; Y: Total Sales, Previous Month Sales	To examine the monthly trend against the previous month
Column chart	X: ProductName (Top 10); Y: Total Sales	To identify the best-selling products
Donut chart	Legend: Country; Values: Total Sales	To identify the countries contributing most to sales
Scatter chart	X: Total Sales; Y: Profit Margin %	To compare sales volume with profitability by category

6. Analysis and Interpretation of Results

- **Overall performance:** The business recorded sales of approximately $24.91M and profit of $10.46M, giving a profit margin of 41.97%. These figures arise from 25,164 orders placed by 17,416 customers, with an average order value of roughly $990.
- **Top-selling products:** The highest-ranked products are all Mountain-200 bikes. Mountain-200 Black 46 leads with about $1.24M, followed by Black 42 (about $1.23M) and Silver 38 (about $1.21M).
- **Category performance:** Bikes generate about $23.64M, or 94.9% of total sales. Accessories contribute roughly $0.91M and Clothing $0.37M. No sales are recorded for Components in this dataset.
- **Profitability:** Bikes account for the highest sales but have the lowest profit margin (41.1%). Accessories yield the highest margin per dollar sold (62.8%), while Clothing occupies an intermediate position (44.3%). High revenue therefore does not necessarily imply the highest margin.

Category	Sales ($)	Profit margin	Share of sales
Bikes	23.64M	41.1%	94.9%
Accessories	0.91M	62.8%	3.6%
Clothing	0.37M	44.3%	1.5%
Components	No sales	-	-

- **Regional performance:** Australia ranks first with about $7.42M, close to 30% of total sales, followed by Southwest (about $4.82M), Northwest (about $3.10M) and the United Kingdom (about $2.90M). Three US regions, namely Central, Northeast and Southeast, record negligible sales of under $12K each.
- **Sales trend:** Annual sales increased from $6.40M in 2020 to $9.32M in 2021. The 2022 data covers January to June only, yet sales had already reached $9.19M, approximately three times the figure for the same six months of 2021. Monthly sales show a consistent upward trend, with June 2022 the strongest month recorded at about $1.83M, an increase of around 3.3% over May.

7. Conclusion

The analysis shows that the sales and profit of Adventure Works are driven overwhelmingly by the Bikes category, led by the Mountain-200 range, and by a small number of markets, with Australia ranking first. Accessories deliver the highest profit margin (62.8%) but account for only 3.6% of sales, which suggests that expanding this category is the most direct means of improving overall margin. Sales exhibit a steady month-on-month increase, and sales for the first half of 2022 are already close to the full-year total for 2021. The near-inactivity of three US regions indicates untapped potential, while the absence of sales for Components warrants verification of the source data.

The twenty-five DAX measures were validated against the raw data, and the dashboard enables these patterns to be explored by year, region and category.